III SEMESTER	L	T	P	C
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20CS3L02 : DATABASE MANAGEMENT SYSTEMS LAB				

COURSE OUTCOMES:

At the end of the course student able to

CO1: Understand the different issues involved in the design and implementation of a database system

CO2: Design entity relationship and convert entity relationship diagrams into RDBMS and formulate SQL queries on the respect data.

CO3: Create and maintain tables using SQL

CO4: Populate and query a database using SQL DML/DDL commands.

CO5: Understand the use of structured query language and its syntax, transactions, database recovery and techniques for query optimization.

CO6: Acquire a good understanding of database systems concepts and to be in a position to use and design databases for different applications.

LIST OF EXPERIMENTS:

1. Create a table STUDENT with appropriate data types and perform the following queries.
   Regdno, student name, date of birth, branch and year of study.
   a. Insert 5 to 10 rows in a table
   b. List all the students of all branches
   c. List student names whose name starts with ‘s’
   d. List student names whose name contains ‘s’ as third literal
   e. List student names whose contains two ‘s’ anywhere in the name
   f. List students whose branch is NULL
   g. List students of AIML who born after 1985
   h. List all students in reverse order of their names
   i. Delete students of any branch whose name starts with ‘s’
   j. Display student name padded with ‘*’ after the name of all the students

2. Create the following tables with appropriate data typesand constraints.
   EMPLOYEE (Fname, Mname, Lname, <u>SSN</u>, Bdate, Address, Gender, Salary, SuperSSN,Dno)
   DEPARTMENT(<u>Dnumber</u>, Dname, MgrSSN, Mgrstartdate)
   DEPENDENT (ESSN, Dependent_Name, Gender, Bdate, Relationship)
   a. Insert 5 to 10 rows into all the tables.
   b. Display all employee’s names along with their department names.
   c. Display all employee’s names along with their dependent details.
   d. Display name and address of all employees who work for ‘AIML’ department.
   e. List the names of all employees with two or more dependents.
   f. List the names of employee who have no dependents.
   g. List the names of employees who have at least one dependent.
   h. List the names of the employees along with names of their supervisors using aliases.
   i. Display name of the department and name of manager for all the departments.
   j. Display the name of each employee who has a dependent with the same first name and gender as the employee.
   k. List the names of managers who have at least one dependent.

l. Display the sum of all employees' salaries as well as maximum, minimum and average salary in the entire departments department wise if the department has more than two employees.
m. List the departments of each female employee along with her name.
n. List all employee names and also the name of the department they manage if they happen to manage a dept

3. Consider the following schema for a Library Database:
BOOK (Book_id, Title, Publisher_Name, Pub_Year)
BOOK_AUTHORS (Book_id, Author_Name)
PUBLISHER (Name, Address, Phone)
BOOK_COPIES (Book_id, Branch_id, No-of_Copies)
BOOK_LENDING (Book_id, Branch_id, Card_No, Date_Out,Due_Date)
LIBRARY_BRANCH (Branch_id, Branch_Name, Address)

Write SQL queries to
a. Retrieve details of all books in the library – id, title, name of publisher, authors, number of copies in each branch
b. Get the particulars of borrowers who have borrowed more than 3 books in particular period.
c. Delete a book in BOOK table. Update the contents of other tables to reflect this data manipulation operation
d. Partition the BOOK table based on year of publication. Demonstrate its working with a simple query
e. Create a view of all books and its number of copies that are currently available in the Library.
f. Apply Drop, Alter, Order by, Group By clauses on the above tables

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4. Create the following tables for university database
Department (deptname, building, budget, primary key (dept name))
Course (courseid, title, deptname, credits, primary key (course id))
Instructor (ID, name, deptname, salary, primary key (ID), foreign key (deptname) references department);
Section(courseid, sec id, semester, year, building, room number, time slot id, primary key (courseid, sec id, semester, year),foreign key (course id) references course);
Teaches (ID, course id ,sec id ,semester ,year numeric ,primary key (ID, course id, sec id, semester, year),foreign key (course id, sec id, semester, year) references section ,foreign key (ID) references instructor);
a. List the names of instructors along with the titles of courses that they teach
b. Find the names of all instructors whose salary is greater than at least one instructor in the AIML department
b. To find the set of all courses taught in the Even as well as in Odd semester
c. Find the average salary of instructors in the Computer Science department
d. Find the average salary in each department
e. To find all the courses taught in the Even semester in current year but not in previous year in the same semester
f. Delete all instructors with a salary between 15,000 and 35,000.

5. Create the following tables based on the above Schema Diagram with appropriate data types and constraints and perform the following queries.
SAILORS (Saild, Salname, Rating, Age)

RESERVES (Sailid, boatid, Day)
BOATS (Boatid, Boat-name, Color)

a. Insert 5 to 10 rows in all tables?
b. Find the name of sailors who reserved boat number 3.
c. Find the name of sailors who reserved green boat.
d. Find the colors of boats reserved by Particular Sailor
e. Find the names of sailors who have reserved at least one boat.
f. Find the all sailid of sailors who have a rating of 10 or have reserved boated 104.
g. Find the Sailid‘s of sailors with age over 20 who have not registered a red boat.
h. Find the names of sailors who have reserved a red or green boat.
i. Find sailors whose rating is better than some sailor called _Salvador‘.
j. Find the names of sailors who are older than the oldest sailor with a rating of 10.

6. Queries on Working with Index, Sequence, Synonym, Controlling Access and Locking Rows for Update, Creating Password and Security features
   a. Write a PL/SQL Block to find whether the number is Armstrong or not.
   b. Write a PL/SQL program for generating Fibonacci series
   c. Write a PL/SQL program to print given number in reverse
   d. print different patterns using PL/SQL program
   e. Print Floyd’s triangle in PL/SQL
   f. Count odd and even digits in a number in PL/SQL

7. Write a program that updates salaries of all employees with 10 % hike (use cursors).
8. Write a program to fetch salary and employee name from employee table for a given user input. When no data found raise an exception that prints the message “no data found”.
9. Write a program to find the number of records of any given table using % ROWCOUNT.
10. Write a database trigger on employee table so that the trigger fires when all the DML statements are executed (print appropriate message).
11. Write a trigger in such a way that it should not allow insert or update or delete on Saturday and Sunday and display the proper message.
12. Write a database trigger on employee table so that the trigger fires when all the DML statements are executed (print appropriate message)
13. Write a cursor to display the list of employees and total salary department wise.
14. Write a procedure to display the name and salary of employee when user inputs SSN using IN/OUT parameters.
15. Write a function to check the validity of the given employee number from the employee table (print the appropriate message using PL/SQL block)
16. Write a PL/SQL Code Creation of forms for any Information System such as Student Information System, Employee Information System etc.